



Guangdong Technion

Israel Institute of Technology

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GTIIT

CALCULUS 1 - M

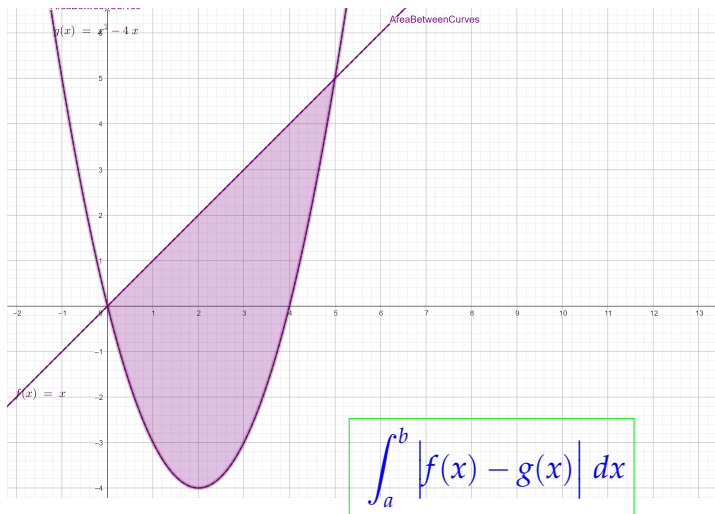
WINTER SEMESTER - 2024

LECTURE 16 - DECEMBER 3RD

- Section 6.1: Areas enclosed by curves.
- Section 6.2: Volumes.
- Section 6.5: Average Value of a function.

AREAS, VOLUMES AND AVERAGE VALUE OF A FUNCTION

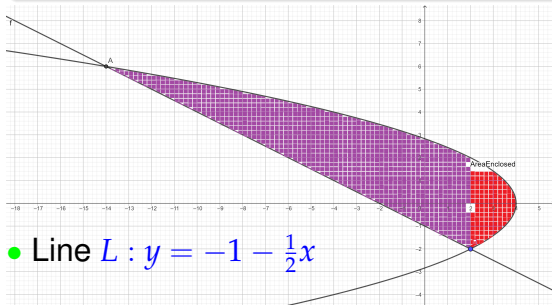
AREA ENCLOSED BY CURVES



=area enclosed by the two curves $y = f(x)$ and $y = g(x)$

AREA ENCLOSED BY CURVES

Ex. 1. Find the area of enclosed by the parabola $y^2 = 8 - 2x$ and the line $2y + 2 = -x$.



• Line $L : y = -1 - \frac{1}{2}x$

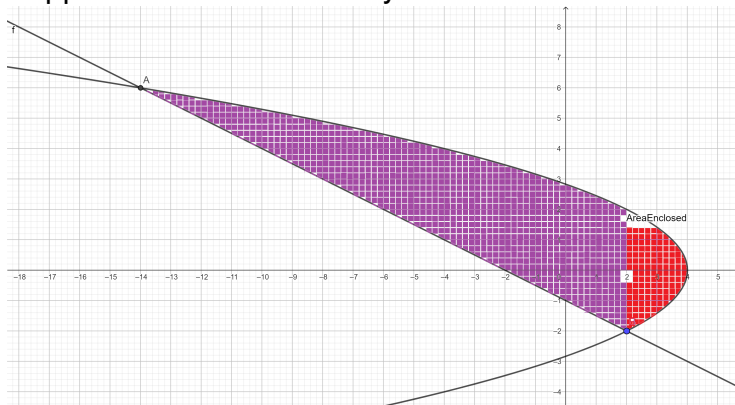
• Find the intersections between the curves:

$$8 - 2x = y^2 = \left(-1 - \frac{1}{2}x\right)^2 \Rightarrow 8 - 2x = \frac{x^2}{4} + x + 1 \Rightarrow x = 2, x = -14$$

• The points $(2, -2)$ and $(-14, 6)$ are the intersections between L and P

AREA ENCLOSED BY THE PARABOLA $P : y^2 = 8 - 2x$ AND THE LINE $L : 2y + 2 = -x$.

- Find upper and lower boundary curves on each interval



- On the left, $-14 \leq x \leq 2 \Rightarrow -1 - \frac{1}{2}x \leq y \leq \sqrt{8 - 2x}$

- On the right, $2 \leq x \leq 4 \Rightarrow -\sqrt{8 - 2x} \leq y \leq \sqrt{8 - 2x}$

The enclosed area is a union of 2 regions with different descriptions: lower and upper boundary curves are not equal in the two intervals.

$$\text{Area} = \int_{-14}^4 \left| f(x) - g(x) \right| dx = A_1 + A_2$$

$$= \int_{-14}^2 \left[\sqrt{8-2x} - (-1 - \tfrac{1}{2}x) \right] dx + \int_2^4 \left[\sqrt{8-2x} - (-\sqrt{8-2x}) \right] dx$$

$$= \int_{-14}^2 \left[\sqrt{8-2x} + 1 + \tfrac{1}{2}x \right] dx + \int_2^4 \left[2\sqrt{8-2x} \right] dx$$

$$= \left[-\frac{1}{3}(8-2x)^{3/2} + x + \frac{1}{4}x^2 \right]_{-14}^2 + \left[-\frac{2}{3}(8-2x)^{3/2} \right]_2^4$$

$$= \frac{208}{3} - 60 + \frac{16}{3} = \frac{128}{3} \quad \checkmark$$

2nd Possibility: integrate with respect to y .

The enclosed area is a region with lower boundary curve given by the line and upper boundary curve given by the parabola.

Define new functions of the form $x = h(y)$ for $-2 \leq y \leq 6$:

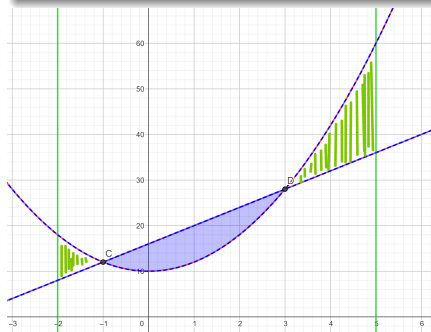
$$h_1(y) = 4 - \frac{1}{2}y^2 \text{ and } h_2(y) = -2y - 2$$

Then

$$\begin{aligned} \text{Area} &= \int_{-2}^6 |h_1(y) - h_2(y)| dy = \int_{-2}^6 \left[\left(4 - \frac{1}{2}y^2\right) - (-2y - 2) \right] dy \\ &= \left[-\frac{1}{6}y^3 + y^2 + 6y \right] \bigg|_{y=-2}^{y=6} \\ &= \frac{128}{3} \checkmark \end{aligned}$$

AREA ENCLOSED BY CURVES

Ex. 2. Find the area of the region enclosed by the curves $y = 2x^2 + 10$, $y = 4x + 16$, $x = -2$ and $x = 5$.



- Line $L : y = -1 - \frac{1}{2}x$
- Find the intersections between the curves:

$$4x + 16 = y = 2x^2 + 10 \Rightarrow 0 = x^2 - 2x - 3 \Rightarrow x = -1, x = 3$$

- The points $(-1, 12)$, $(3, 28)$ are the intersections between L and the parabola

The enclosed area is a union of 3 regions with different descriptions.

The **upper and lower** boundary curves interchange their rolls in each of the three intervals, so, the area is a sum:

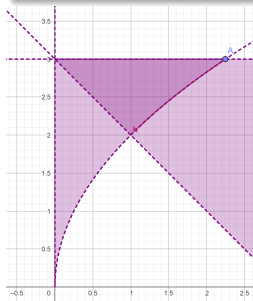
$$\text{Area} = A_1 + A_2 + A_3$$

$$\begin{aligned} &= \int_{-2}^{-1} \left[\underbrace{2x^2 + 10}_{\text{upper bound}} - \underbrace{(4x + 16)}_{\text{lower bound}} \right] dx + \int_{-1}^3 \left[\underbrace{4x + 16}_{\text{upper bound}} - \underbrace{(2x^2 + 10)}_{\text{lower bound}} \right] dx \\ &\quad + \int_3^5 \left[\underbrace{2x^2 + 10}_{\text{upper bound}} - \underbrace{(4x + 16)}_{\text{lower bound}} \right] dx \\ &= \left[2x^2 - 4x - 6 \right]_{-2}^{-1} + \left[-2x^2 + 4x + 6 \right]_{-1}^3 + \left[2x^2 - 4x - 6 \right]_3^5 \end{aligned}$$

$$= \frac{14}{3} + \frac{64}{3} + \frac{64}{3} = \frac{142}{3} \quad \checkmark$$

Ex. 3. Find the area of the region enclosed by the curves

$y^2 = 4x$, $y = 3 - x$ and $y = 3$ in the first quadrant.



- Intersection between the horizontal line and the parabola:

$$3 = y = 2\sqrt{x} \Rightarrow x = \frac{9}{4} \Rightarrow A = \left(\frac{9}{4}, 3\right)$$

- Intersection between the slant line and the parabola:

$$3 - x = y = 2\sqrt{x} \Rightarrow x = 1, x = 9$$

But we only need $B = (1, 2)$ in the 1st quadrant

- Intersection between the two lines: $C = (0, 3)$

The enclosed area is a union of 2 regions with different lower boundary curves.

$$\text{Area} = A_1 + A_2$$

$$\text{Area} = \int_a^b |f(x) - g(x)| dx$$

$$= \int_0^1 [\text{Horizontal} - \text{Slant line}] dx + \int_1^{\frac{9}{4}} [\text{Horizontal line} - \text{Parabola}] dx$$

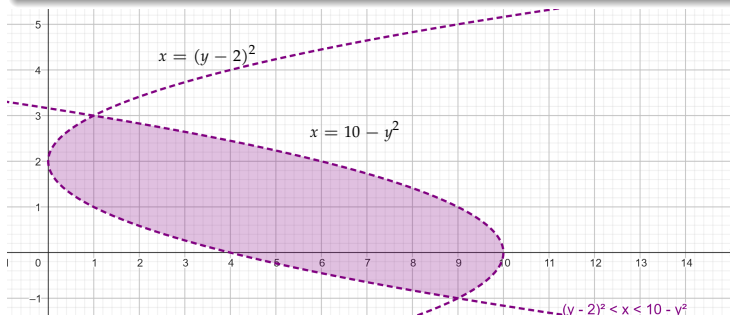
$$= \int_0^1 [3 - (3 - x)] dx + \int_1^{\frac{9}{4}} [3 - 2\sqrt{x}] dx$$

$$= \int_0^1 x dx + 3 \cdot \frac{5}{4} - \int_1^{\frac{9}{4}} 2\sqrt{x} dx$$

$$= \frac{1}{2} + \frac{15}{4} - \frac{4}{3} x^{3/2} \Big|_1^{\frac{9}{4}} dx = \frac{1}{2} + \frac{15}{4} - \frac{19}{6} = \frac{13}{12} \checkmark$$

AREA ENCLOSED BY TWO CURVES

Ex. 4. Find the area enclosed by the parabolas $x = 10 - y^2$ and $x = (y - 2)^2$.



- Find the intersections between the curves:

$$10 - y^2 = x = (y - 2)^2 \Rightarrow 0 = 2y^2 - 4y - 6 \Rightarrow y = -1, y = 3$$

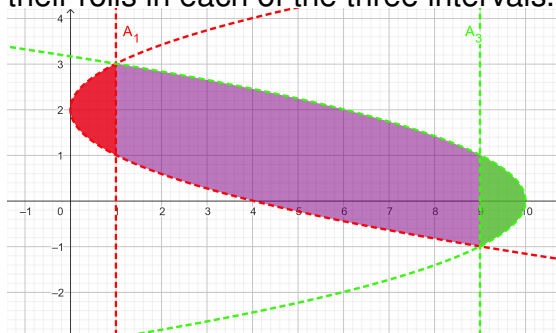
- $(1, 3)$, $(9, -1)$ are the intersections points

INTEGRATE WITH RESPECT TO y

- Integrating with respect to the variable x requires splitting into a sum of 3 integrals

$$\text{Area} = A_1 + A_2 + A_3$$

since the upper and lower boundary curves interchange their rolls in each of the three intervals.



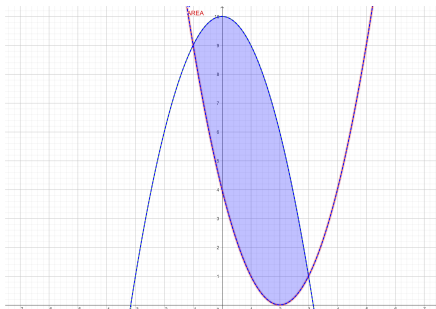
- Instead, we may integrate with respect to the variable y

INTEGRATE WITH RESPECTO TO y

$$\text{Area} = \int_{-1}^3 \left[\left(\text{right function} \right) - \left(\text{left function} \right) \right] dy$$

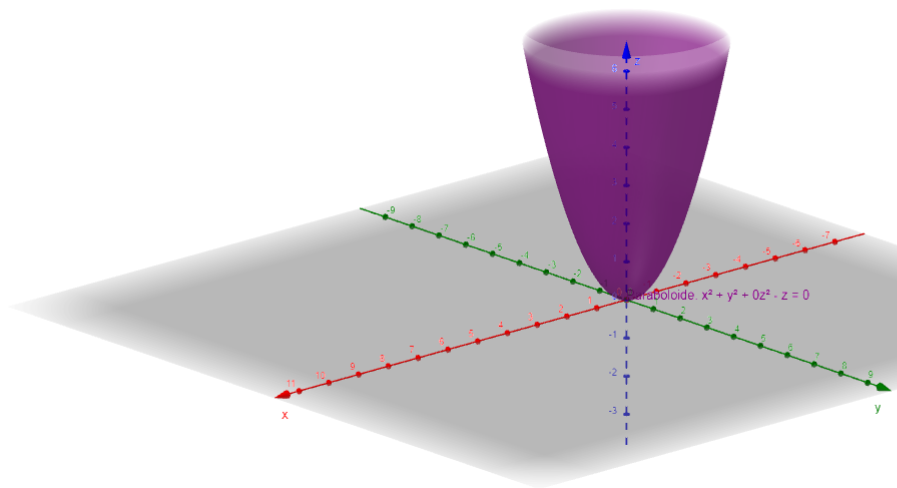
$$\int_{-1}^3 \left[\left(10 - y^2 \right) - \left(y - 2 \right)^2 \right] dy$$

$$\int_{-1}^3 \left[-2y^2 + 4y + 6 \right] dy = \left[-\frac{2}{3}y^3 + 2y^2 + 6y \right]_{-1}^3 = \frac{64}{3} \checkmark$$



THE SPACE $\mathbb{R}^3$

Consider $\mathbb{R}^3$, the 3-dimensional space.



COMPUTE THE VOLUMEN OF A SOLID IN THE SPACE $\mathbb{R}^3$

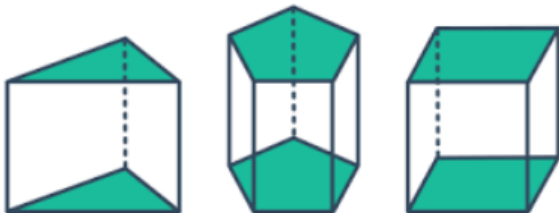
- A **solid** is a set contained in **the space $\mathbb{R}^3$** , bounded by surfaces
- Consider a simple type of solid, enclosed by a **right cylinder**, with parallel base and top, B_1 and B_2

- **Solid bounded by a right cylinder:**

$$= \{(x, y, z) \in \text{segments, which are perpendicular to } B_1, \\ \text{joining } B_1 \text{ and } B_2\}$$

- A **cylinder** can be a rectangular box, a circular cylinder, or one with a base of any form.

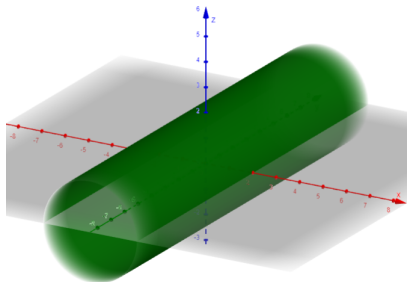
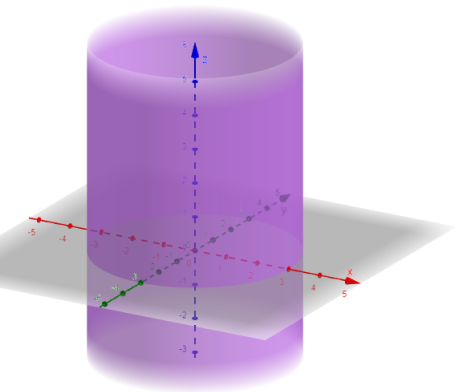
Triangular, Pentagonal, Rectangular Cylinders:



$$\bullet V_1 = \frac{1}{2}\ell_1\ell_2h, \quad V_2 = \frac{\sqrt{25+10\sqrt{5}}}{4}\ell^2h, \quad V_3 = \ell_1\ell_2h$$

- *Volume* = area of the base $\times h$
- Volume has dimensions of $[\text{units of length}]^3$
- Triangular, Pentagonal, Rectangular Cylinders or with any kind of base

SOLID BOUNDED BY A CIRCULAR CYLINDER

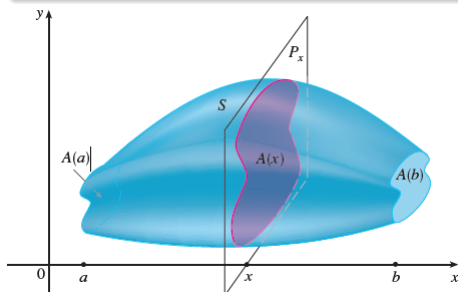


$$\text{Volume} = \text{area of the base} \times h \quad \Rightarrow V = \pi R^2 h$$

COMPUTE A VOLUME FOR A GENERAL SOLID: USING INTEGRALS

- For a solid S that is not a cylinder, cut S into thin slices: We start by intersecting S with a plane, **parallel to one of the 3 coordinate planes**, and obtaining a plane region that is called a **cross-section** of S .
- **Case 1:** take $A(x)$ the **area of the cross-section of S in a plane that is parallel to the y, z plane, perpendicular to the x -axis and passing through the point x** , where $a < x < b$.
- Think of slicing S through x and computing the area of this slice. The cross-sectional area $A(x)$ varies as x increases from a to b .
- Instead of multiplying by the height h , **integrate with respect to x on $[a, b]$**

Definition of Volume



- Let S be a solid that lies between the planes $x = a$ and $x = b$.
- Let $A(x)$ be the area of the cross-section of S passing through the point x , in a perpendicular plane to the x -axis, where $a < x < b$.
- If $A(x)$ is a continuous function, then the volume of S is

$$V = \int_a^b A(x) dx$$

REVOLUTION SOLIDS

A **Revolution Solid** is obtained by the rotation of a region, enclosed by curves, with respect to a given axis: In this case, each slice, perpendicular to the axis, is a **circle**.

- If the region lies below a curve $y = f(x)$ on $[a, b]$ and the rotation is around **the x - axis**, the area function is

$A(x) = \pi [f(x)]^2$ and the volume is

$$V = \int_a^b \pi [f(x)]^2 dx$$

- If the region lies below a curve $x = g(y)$ on $[c, d]$ and the rotation is around **the y - axis**, the area function is

$A(y) = \pi [g(y)]^2$ and the volume is

$$V = \int_c^d \pi [g(y)]^2 dy$$

The rotation of a region, enclosed by curves, makes a **surface**:
the **revolution solid** is the content, enclosed and bounded by this surface.
Go the Geogebra link to see the animation on **Revolution Solids**:

Revolution Solids: $y = x$

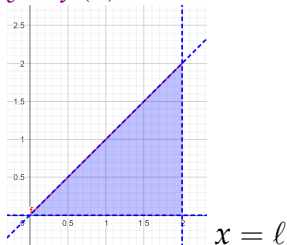
Revolution Solids: $y = \sqrt{x}$

REVOLUTION SOLIDS

Ex. 1. Compute the volume of the solid obtained by the rotation of the curve $y = x$ around the x -axis on $[0, \ell]$, where $\ell > 0$.

Solution. • Each vertical slice, perpendicular to the x -axis, is a circle of radius $f(x) = x$

$$y = f(x) = x$$



Rotation around the x -axis

VOLUME OF THE CONE

- Any vertical slice on a perpendicular plane is a circle of radius $y = f(x) = x$.
- The cross-sectional **area of the circle of radius $f(x)$** is the continuous function

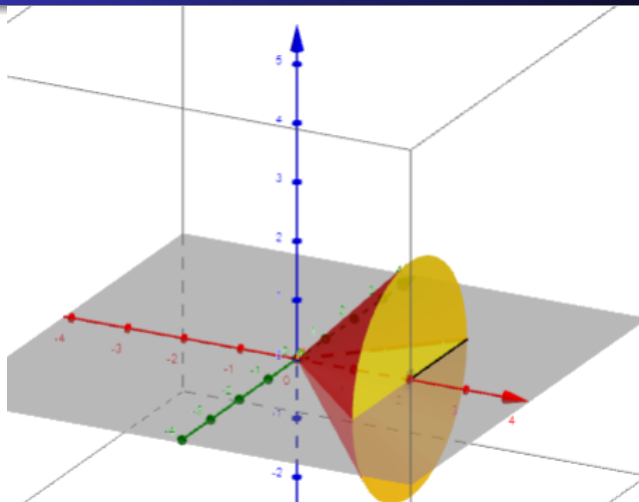
$$A(x) = \pi \times \text{radius}^2 = \pi x^2$$

- The volume** is the integral of the **area**

$$V = \int_0^{\ell} A(x) \, dx = \int_0^{\ell} [\pi x^2] \, dx$$

$$= \left. \frac{\pi}{3} x^3 \right|_0^{\ell} = \frac{\ell^3}{3} \pi \checkmark$$

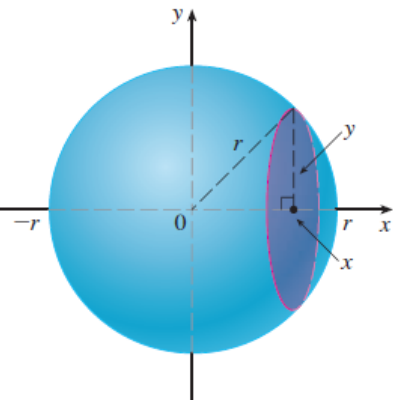
VOLUME OF THE CONE



$$V = \frac{\ell^3}{3}\pi \checkmark$$

REVOLUTION SOLIDS

Ex. 2. Compute the volume of the solid sphere of radius r

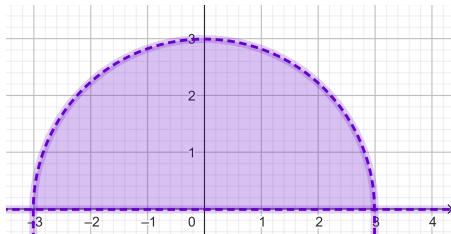


$$y = f(x) = \sqrt{r^2 - x^2}$$

Solution. • If we place the sphere so that its center is at the origin, then **the sphere lies between the vertical planes $x = -r$ and $x = r$.**

THE SPHERE AS A REVOLUTION SOLID: CURVE

$$y = f(x) = \sqrt{r^2 - x^2}$$



- $x^2 + y^2 = r^2 \Rightarrow y = f(x) = \sqrt{r^2 - x^2}$
- $y > 0$ on the upper half-plane

VOLUME OF THE SOLID SPHERE

- The points on the surface of the sphere verify $x^2 + y^2 = r^2$.
 - Any vertical slice on a perpendicular plane is a circle whose radius, **from the Pythagorean Theorem**, is

$$y = \sqrt{r^2 - x^2} : y \geq 0$$

- The cross-sectional area is

$$A(x) = \left[\sqrt{r^2 - x^2} \right]^2 = \pi(r^2 - x^2)$$

- The volume** is the integral of the **even**, continuous function $\pi[A(x)]^2$:

$$V = \int_{-r}^r A(x) dx = \int_{-r}^r \pi(r^2 - x^2) dx$$

$$= 2\pi \int_0^r [r^2 - x^2] dx = 2\pi \left[r^2 x - \frac{1}{3} x^3 \right]_0^r = \frac{4}{3} \pi r^3 \checkmark$$

REVOLUTION VOLUMES

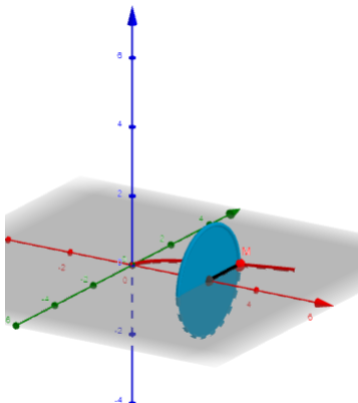
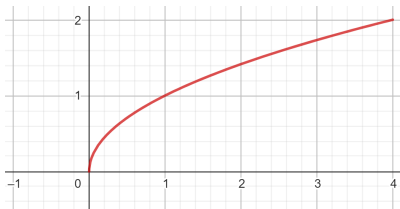
Ex. 3. Find the volume of the solid obtained by revolution of $y = \sqrt{x}$ with respect to the x - axis on $[0, 4]$.

- Radius of each slice

$$f(x) = \sqrt{x}$$

- Area of each slice

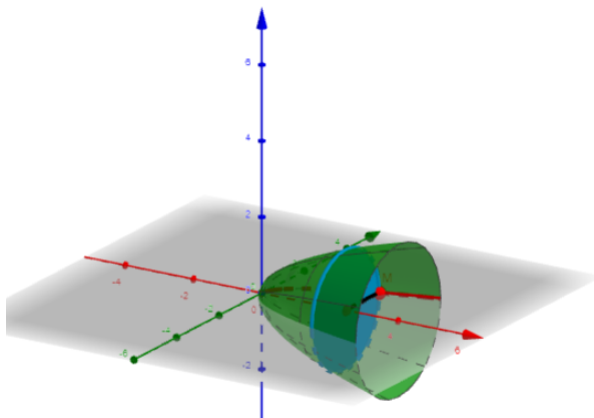
$$= \pi[f(x)]^2 = \pi x$$



REVOLUTION VOLUMES

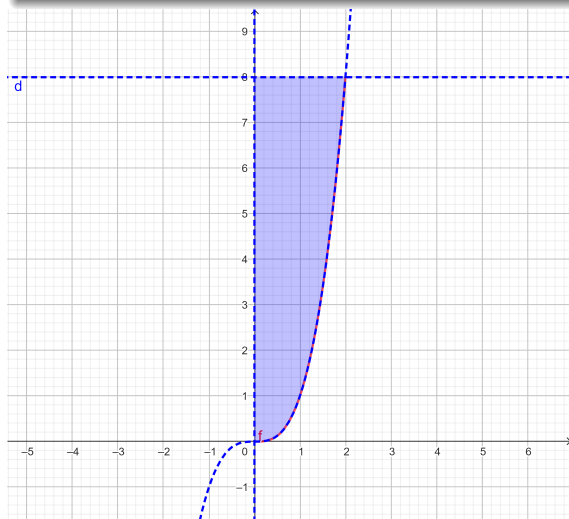
- The volume of the solid obtained by revolution of $y = \sqrt{x}$ with respect to the x - *axis* on $[0, 4]$ is then

$$V = \int_0^4 \pi [f(x)]^2 dx = \int_0^4 \pi x dx = \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$



REVOLUTION SOLIDS

Example 4. Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$ and $x = 0$ about the y -axis.



Solution. • Because the region is obtained by a rotation about the **y-axis**, it makes sense to slice the solid perpendicular to the **y-axis**, obtaining circular cross-sections.

• For $y \in [0, 8]$, if we slice at height y , we get a circular disk with radius $x = g(y)$:

$$y = x^3 \iff x = \sqrt[3]{y} : 0 \leq y \leq 8$$

• The **cross-sectional area is a function on y** , therefore, integrate **with respect to y** :

$$\Rightarrow V = \int_0^8 A(y) dy = \int_0^8 \pi [g(y)]^2 dy = \int_0^8 \pi y^{2/3} dy$$

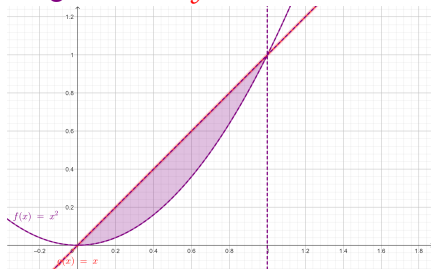
$$= \frac{3}{5} \pi y^{5/3} \Big|_0^8 = \frac{96\pi}{5} \checkmark$$

REVOLUTION SOLIDS

Example 5. a) Consider the region $\mathcal{R}$ enclosed by the curves $y = x$ and $y = x^2$.

If this region is rotated around the x -axis, find the volume of the resulting solid.

Region $\mathcal{R}$ $y = x$



$y = x^2$

$$0 \leq x^2 \leq x \leq 1$$

The parabola is below the line on $[0, 1]$

REVOLUTION SOLIDS: ANNULUS

Solution. • The curves $y = x$ and $y = x^2$ intersect at the points $x = 0$ and $x = 1$.

- Consider the rotation around the x -axis, of the region between the curves.
- Next, find the area of a cross-section, perpendicular to the x -axis.
- A cross-section in the plane x, y has the shape of an **annular ring**, with **inner radius** $r = x^2$ and **outer radius** $r = x$, because:

$$0 \leq x^2 \leq x \leq 1$$

REVOLUTION SOLIDS

- We find the cross-sectional area by **subtracting the area of the inner circle from the area of the outer circle**:

$$A(x) = \pi x^2 - \pi x^4$$

Therefore we have

$$\begin{aligned} V &= \int_0^1 A(x) dx = \int_0^1 \pi [x^2 - x^4] dx = \\ &= \pi [x^2 - x^4] dx \Big|_0^1 = \pi \left[\frac{1}{3} x^3 - \frac{1}{5} x^5 \right]_0^1 = \frac{2\pi}{15} \checkmark \end{aligned}$$

REVOLUTION SOLIDS

Example 5. b) Consider the region $\mathcal{R}$ enclosed by the curves $y = x$ and $y = x^2$.

If this region is rotated around **the line $y = 2$** , find the volume of the resulting solid.

Solution: the inner and outer radius are the **distances from the inner and outer curve, respectively, to the line $y = 2$**

REVOLUTION SOLIDS: ANNULUS



ROTATION OF THE ANNULUS AROUND THE LINE $y = 2$

Solution. • The curves $y = x$ and $y = x^2$ intersect at the points $x = 0$ and $x = 1$.

- Compute the area of a cross-section, perpendicular to the line $y = 2$
- A cross-section parallel to the plane x, z (as before) has the shape of an annular ring, with
 - inner radius $y = 2 - x$, and
 - outer radius: $y = 2 - x^2$

REVOLUTION SOLIDS

- We find the cross-sectional area by subtracting the area of the inner circle from the area of the outer circle:

$$\text{Difference of areas} = A(x) = \pi(2 - x^2)^2 - \pi(2 - x)^2$$

Therefore, the volume is the integral of $A(x)$ between the intersection points:

$$\begin{aligned} V &= \int_0^1 A(x) dx = \int_0^1 \pi \left[(2 - x^2)^2 - (2 - x)^2 \right] dx = \pi \int_0^1 [x^4 - 5x^2 + 4x] dx \\ &= \pi \left[\frac{1}{5}x^5 - \frac{5}{3}x^3 + 2x^2 \right] \bigg|_0^1 = \frac{8\pi}{15} \checkmark \end{aligned}$$

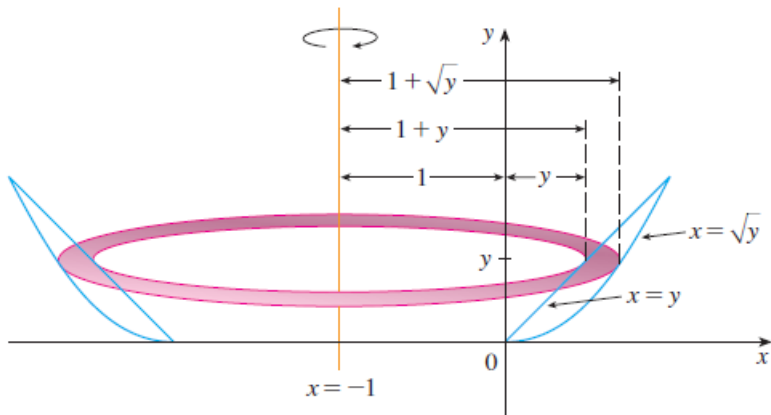
REVOLUTION SOLIDS

Example 5. c) Consider the region $\mathcal{R}$ enclosed by the curves $y = x$ and $y = x^2$ and the rotation of this region around **the line $x = -1$** . Find the volume of the resulting solid.

Solution:

- The inner and outer radius are the **distances from the inner and outer curve, respectively, to the line $x = -1$**
- **Integrate with respect to y** , since the rotation line is parallel to the y -axis.

REVOLUTION SOLIDS: ANNULUS



- Outer radius: $g(y) = x = |-1 - \sqrt{y}| = 1 + \sqrt{y}$
- Inner radius $h(y) = x = |-1 - y| = 1 + y$,

ROTATION AROUND THE LINE $x = -1$

- The curves $y = x$ and $y = x^2$ intersect at the points $x = 0$ and $x = 1$.
- Compute the area of a cross-section, perpendicular to the line $x = -1$
- A cross-section parallel to the plane y, z has the shape of an annular ring, with
 - outer radius: $g(y) = x = \left| -1 - \sqrt{y} \right| = 1 + \sqrt{y}$
 - inner radius $h(y) = x = \left| -1 - y \right| = 1 + y,$

ROTATION AROUND THE LINE $x = -1$

- We find the cross-sectional area by subtracting the area of the inner circle from the area of the outer circle:

$$\text{Difference of areas} = A(y) = \pi(1 + \sqrt{y})^2 - \pi((1 + y))^2$$

- Therefore, the volume is the integral of $A(y)$ between the intersection points:

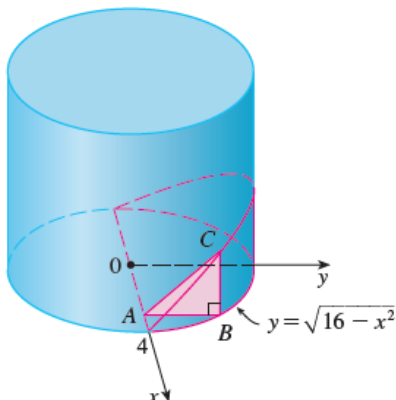
$$V = \int_0^1 A(y) dy = \int_0^1 \pi \left| [g(y)]^2 - [h(y)]^2 \right| dy$$

$$V = \int_0^1 \pi \left[(1 + \sqrt{y})^2 - (1 + y)^2 \right] dy = \pi \int_0^1 [2\sqrt{y} - y - y^2] dy$$

$$= \pi \left[\frac{4}{3}y^{3/2} - \frac{1}{2}y^2 - \frac{1}{3}y^3 \right] \Big|_0^1 = \frac{\pi}{2} \checkmark$$

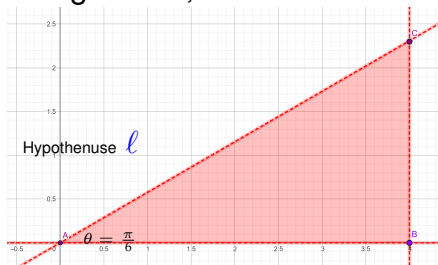
SOLIDS THAT ARE NOT SOLIDS OF REVOLUTION

Ex. 6. Consider a circular cylinder of radius 4, with base on the plane x, y such that the z -axis is its symmetry axis. A wedge is cut out of the cylinder by two planes: the plane x, y and a second plane, that intersects the plane x, y at an angle of 30° , along a diameter of the cylinder on the x -axis. Find the volume of the wedge.



Solution.

- The base of the cylinder is the circle on the plane x, y of all points such that verify $x^2 + y^2 = 16$: $-4 \leq x \leq 4$
- A cross-section at a distance x from the origin is a triangle ABC , whose base and height are:



- Base = $AB = \ell \cos\left(\frac{\pi}{6}\right) = \sqrt{16 - x^2}$,
- Height = $BC = \ell \sin\left(\frac{\pi}{6}\right) = \ell \tan\left(\frac{\pi}{6}\right) \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{16 - x^2}}{\sqrt{3}}$

VOLUME OF THE WEDGE

- The area of a cross-section is equal to the **area of the triangle** $= \frac{1}{2} \text{ base} \times \text{height}$

$$= A(x) = \frac{1}{2} \sqrt{16 - x^2} \frac{\sqrt{16 - x^2}}{\sqrt{3}} = \frac{16 - x^2}{2\sqrt{3}}$$

Therefore, the **volume** is the integral of $A(x)$ on $[-4, 4]$

$$V = \int_{-4}^4 A(x) dx = \int_{-4}^4 \left[\frac{16 - x^2}{2\sqrt{3}} \right] dx = \frac{1}{\sqrt{3}} \int_0^4 [16 - x^2] dx$$

$$= \frac{1}{\sqrt{3}} \left[16x - \frac{1}{3}x^3 \right] \Big|_0^4 = \frac{4^3}{\sqrt{3}} \frac{2}{3} = \frac{128}{3\sqrt{3}} \checkmark$$

AVERAGE VALUE OF A FUNCTION

- The **average** of two numbers is $A = \frac{a + b}{2}$
- The **average** of n numbers $y_1 + \cdots + y_n$ is

$$y_{average} = \frac{y_1 + \cdots + y_n}{n}$$

- Suppose that $T = f(x)$ represents the temperature function on a day and there are temperature readings every hour:

$$f(x_1^*), \dots, f(x_{24}^*)$$

Then, for $n = 24$, the **average temperature on a day** is

$$T_{average} = \frac{f(x_1^*) + \cdots + f(x_{24}^*)}{24}$$

AVERAGE VALUE OF A FUNCTION

In general, let's try to compute the **average value of a function** $y = f(x)$ on $[a, b]$.

- We start by dividing the interval $[a, b]$ into n equal subintervals of length $\Delta x = \frac{(b-a)}{n}$
- Choose points $x_1^*, \dots, x_n^*$ in successive subintervals $[x_1, x_2], \dots, [x_n, x_{n+1}]$
- Compute the **average of the numbers** $f(x_1^*), \dots, f(x_n^*)$:

$$\text{Average of } f\text{-readings} = \frac{f(x_1^*) + \dots + f(x_n^*)}{n} = \frac{f(x_1^*) + \dots + f(x_n^*)}{\frac{(b-a)}{\Delta x}}$$

$$= \frac{1}{(b-a)} \sum_{k=1}^n f(x_k^*) \Delta x$$

- If we let n increase, we compute the average value of a large number of closely spaced values: for example, we would be averaging temperature readings taken every

AVERAGE VALUE OF f ON $[a, b]$

- Take limit of the **Riemann Sums** as $n \rightarrow \infty$:

$$f_{average} = \lim_{n \rightarrow \infty} \frac{1}{(b-a)} \sum_{k=1}^n f(x_k^*) \Delta x$$

- The limit value is a definite integral by the definition.

The Average of f on $[a, b]$ is

$$f_{average} = \frac{1}{(b-a)} \int_a^b f(x) dx$$

AVERAGE VALUE OF f ON $[a, b]$

Ex. 7. a) Find the average value of the function $f(x) = 1 + x^2$ on the interval $[-1, 2]$.

Solution. The Average of f on $[a, b]$ is

$$f_{\text{average}} = \frac{1}{(b-a)} \int_a^b f(x) dx$$

$$= \frac{1}{3} \int_{-1}^2 [1 + x^2] dx$$

$$= \frac{1}{3} \left[x + \frac{1}{3}x^3 \right] \Big|_{-1}^2 = \frac{1}{3} \left[\left(2 + \frac{1}{3}2^3 \right) - \left(-1 + \frac{1}{3}(-1)^3 \right) \right] \Big|_{-1}^2 = 2$$

MEAN VALUE THEOREM FOR INTEGRALS

- If f is the temperature at time x , we might wonder if there is a specific time when the temperature is the same as the average temperature.
- In general, is there a number $c \in [a, b]$ at which the value $f(c)$ is exactly equal to the average value of the function, that is, $f(c) = f_{\text{average}}$?
- The Mean Value Theorem for Integrals says that this is true for continuous functions.

MEAN VALUE THEOREM FOR INTEGRALS

If f is continuous on $[a, b]$, then there exists $c \in [a, b]$ such that

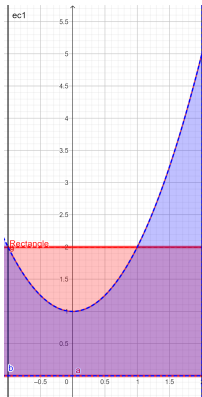
$$f(c) = f_{\text{average}} = \frac{1}{(b-a)} \int_a^b f(x) \, dx$$

Equivalently, there exists $c \in [a, b]$ such that

$$\int_a^b f(x) \, dx = f(c)(b-a)$$

THE MEAN VALUE THEOREM FOR INTEGRALS

The geometric interpretation is that, for **positive, continuous functions f** , there is $c \in [a, b]$ such that the **rectangle** with base $[a, b]$ and height $f(c)$ has the **same area** as the region under the graph of f from a to b .



$$\int_a^b f(x) dx = f(c)(b - a)$$

$$\text{Average Height} = \frac{\text{area}}{\text{width}}$$

THE MEAN VALUE THEOREM FOR INTEGRALS

Ex. 7. b) For the function of the previous exercise, $f(x) = 1 + x^2$, find c such that $f(c) = f_{\text{average}}$.

In this case, we look for $c \in [-1, 2]$ such that

$$1 + c^2 = f(c) = f_{\text{average}} = \frac{1}{3} \int_{-1}^2 [1 + x^2] dx = \frac{1}{3} \int_{-1}^2 [1 + x^2] dx = 2$$

$$\iff c^2 = 1 \iff c = -1 \text{ and } c = 1 \in [-1, 2]$$

So, for these two values:

$$f(-1) = f_{\text{average}} = \frac{1}{3} \int_{-1}^2 [1 + x^2] dx = 2 = f(1)$$

The area under the light-green curve is the same as the area of the violet rectangle

$$f(-1) = f_{average} = \frac{1}{3} \int_{-1}^2 [1 + x^2] dx = 2 = f(1)$$

